UNIT - 5

APPLICATIONS OF NANO TECHNOLOGY

Materials - Biology & medicine - Surface science - Energy & environment - Thin films - Quantum dots

1. Overview of Applications

Because of the large surface-to-volume ratio and quantum effects, nanomaterials give better strength, reactivity, conductivity, optical response and selectivity. Their applications cut across every field:

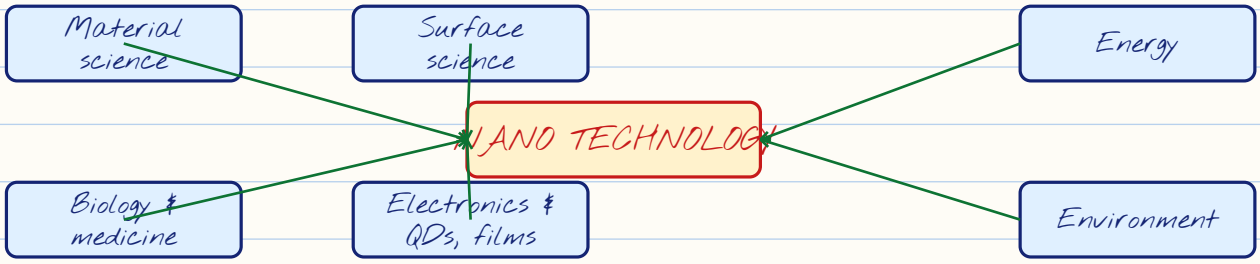


Fig: Fields where nanotechnology is applied

2. Applications in Material Science

2.1 Nanostructured and nanocrystalline materials

- **Nanocrystalline metals and alloys**: grain size 5 - 100 nm gives 2 - 10 times higher hardness / strength (Hall-Petch), better wear and fatigue resistance. Used in cutting tools, gears, bearings, coatings, high-strength steels, Ni-W / Ni-Co electrodeposits
- **Nanoceramics**: nano ZrO_2, Al_2O_3, SiC, Si_3N_4 are tougher, superplastic and sinter at lower temperature (transparent ceramics for armour, bone implants, engine parts, thermal barrier coatings)
- **Hard coatings**: nanocomposite TiN / SiN_x, TiAlN, nc-TiC / a-C with hardness > 40 GPa (superhard), cermets WC-Co with nano WC for drills
- **Magnetic materials**: nanocrystalline Fe-Si-B (Finemet) soft magnets for transformers, Nd-Fe-B nano-composite permanent magnets (exchange-coupled), ferrofluids for seals and speakers

2.2 Nanocomposites

A nanocomposite = matrix + filler where at least one dimension of the filler is < 100 nm.

Type	Examples	Benefits
Polymer matrix (PNC)	Nylon + nanoclay (Toyota timing belt cover), epoxy + CNT / graphene, PET + SiO_2, rubber + carbon black / silica in tyres	Higher strength and modulus at 1 – 5 wt %, barrier to gas (packaging), flame retardancy, conductivity
Ceramic matrix (CMC)	Al_2O_3 + SiC nanoparticles / Al_2O_3 + ZrO_2	Improved fracture toughness, creep and wear resistance
Metal matrix (MMC)	Al + CNT, Cu + graphene, Mg + SiC, Ni + Al_2O_3	High strength to weight, thermal / electrical conductivity

2.3 Smart and functional coatings

- **Self-cleaning** glass (TiO_2: photocatalytic + hydrophilic – decomposes dirt under UV, rain washes it off) and **Lotus effect** superhydrophobic paints (contact angle > 150°)
- **Anti-reflection and anti-fog** coatings (nano porous SiO_2) for lenses and solar panels; **UV blocking** using nano ZnO / TiO_2 in sunscreens, paints, fabrics (transparent as particle < wavelength)
- **Anti-corrosion, anti-scratch** coatings for automobiles, and **anti-bacterial** surfaces with nano Ag, fire retardant nano-silica gels
- **Electrochromic smart windows** (WO_3 films) and thermochromic VO_2 coatings

2.4 Textiles, construction and electronics

- Stain-resistant, wrinkle-free and anti-odour fabrics (nano-whiskers, Ag particles); conductive smart textiles
- Nano-silica and nano-TiO_2 in cement: higher strength, self-cleaning façades, air purifying roads; aerogel insulation
- **Electronics**: shrinking transistor (7 nm / 5 nm node), high-K gate

dielectrics (HfO_2), nano-flash memory, spintronics (GMR, TMR heads), MEMS / NEMS sensors, conductive inks and printed circuits, quantum computing devices

- Nano-lubricants (MoS_2, WS_2, fullerene-like nanoparticles) cut friction; nano-catalysts in refineries; nano-abrasives for polishing (CMP slurry)

3. Applications in Biology and Medicine (Nanomedicine)

Biological units (DNA 2 nm, proteins 5 – 10 nm, viruses 20 – 300 nm, cell organelles) are on the nano scale, so nanoparticles can interact with them directly and can enter cells.

3.1 Drug delivery

- Nanocarriers: **liposomes**, polymeric nanoparticles (PLGA), dendrimers, micelles, gold / silica / magnetic nanoparticles, CNT, nano-gels, albumin-bound (Abraxane)
- **Passive targeting – EPR effect**: tumour blood vessels are leaky (gaps 100 – 800 nm), so 10 – 200 nm particles accumulate in tumours and are retained (poor lymphatic drainage)
- **Active targeting**: surface ligands (antibodies, folic acid, aptamers, peptides RGD) bind to receptors overexpressed on cancer cells. PEGylation ("stealth") prolongs circulation by avoiding immune capture
- **Controlled / triggered release** by pH (acidic tumour 6.5, endosome 5), temperature, enzymes, light, ultrasound or magnetic field
- Benefits: reduces dosage and side effects, protects drug from degradation, crosses the blood-brain barrier, improves solubility of hydrophobic drugs. Examples: Doxil (liposomal doxorubicin), Onpattro (siRNA lipid NP), Pfizer / Moderna COVID-19 mRNA vaccines in lipid nanoparticles

3.2 Diagnosis and imaging

- **Quantum dots** (CdSe/ZnS, InP, carbon dots) as photostable bright fluorescent labels for cells, tissues and in-vivo tumour imaging, multiplexed detection

- **Magnetic nanoparticles** (Fe_3O_4, SPIONs) as MRI T_2 contrast agents (Feridex) and for magnetic separation of cells / DNA; Gd-based nano contrast for T_1 MRI
- **Gold nanoparticles**: lateral flow immunoassay (pregnancy test and COVID antigen strips - red colour from SPR), colorimetric detection of DNA, SERS probes, CT contrast
- **Biosensors and lab-on-a-chip**: nanowire FET sensors, CNT / graphene glucose sensors, nano-pore DNA sequencing (Oxford Nanopore), nanocantilever sensors for proteins, microfluidic point-of-care diagnostics
- **Nano-diagnostics** for early cancer markers (PSA, circulating tumour cells), photoacoustic imaging

3.3 Therapy

- **Photothermal therapy (PTT)**: Au nanorods / nanoshells absorb near-IR (700 - 900 nm, tissue window) and heat tumour to 42 - 50 °C, killing cells. **Photodynamic therapy (PDT)**: nanoparticle photosensitiser generates singlet oxygen
- **Magnetic hyperthermia**: Fe_3O_4 particles heated in an alternating magnetic field (superparamagnetic loss) within tumour (NanoTherm therapy for glioblastoma)
- **Gene therapy**: lipid / polymer nanoparticles carry DNA, mRNA, siRNA, CRISPR into cells; **immunotherapy** nano-vaccines
- **Antimicrobial** nano-silver and copper (wound dressings, catheter coatings, masks), nano-ZnO against resistant bacteria
- **Tissue engineering**: nano-fibre scaffolds (electrospun PCL, collagen) mimic extracellular matrix; nano-hydroxyapatite for bone and dental implants; nano-diamond coatings, titanium nanotube implants; nerve regeneration guides; artificial skin
- **Nanorobots / nano-machines** (concept): DNA origami robots, micro-swimmers for targeted delivery; "theranostics" combine therapy + diagnostics in one particle

Particle	Size	Medical use
Liposome	50 - 200 nm	Drug and vaccine delivery

Gold nanoparticle / nanorod	5 – 100 nm	Diagnostics, PTT, imaging
SPION (Fe_3O_4)	5 – 20 nm	MRI, hyperthermia, magnetic targeting
Quantum dot	2 – 10 nm	Fluorescence imaging, labelling
Dendrimer	2 – 10 nm	Drug and gene carriers
Carbon nanotube / graphene oxide	1 nm dia., 1 µm length	Biosensing, delivery, PTT

Caution:
Nanotoxicity: size, shape, surface charge and coating decide biodistribution, clearance (Kidney clears < 6 nm, larger particles go to liver and spleen) and possible toxicity. Regulation and safety studies are essential before clinical use.

4. Applications in Surface Science

Nanostructuring a surface changes its energy, wettability, reactivity and adhesion. Surface science with nano techniques controls what happens at interfaces.

4.1 Wettability – superhydrophobic and superhydrophilic surfaces

Young: $\cos\theta = (\gamma_{SV} - \gamma_{SL}) / \gamma_{LV}$ Cassie: $\cos\theta^* = f(\cos\theta + 1) - 1$

- Surface roughness at nano / micro scale (lotus leaf: papillae 10 µm covered with wax nano-crystals) traps air (Cassie-Baxter state) so water beads up with $\theta > 150°$ and rolls off with sliding angle < 10° taking dirt away (lotus effect)
- Applications: self-cleaning windows, paints, textiles, anti-icing aircraft wings, anti-fouling ship hulls, oil-water separation meshes, drag reduction, waterproof electronics (liquid repellent nano coating)
- **Superhydrophilic TiO₂** ($\theta \sim 0°$ under UV): anti-fogging mirrors, easily washed
- **Gecko effect**: millions of nano-hairs (setae, spatulae ~ 200 nm)

stick by van der Waals force → dry adhesives, climbing robots

- **Shark skin** riblets reduce drag and bacterial growth. **Moth eye** nano pillars give anti-reflection

4.2 Catalysis and surface reactivity

- Nano-catalysts have a huge number of surface and edge atoms: Au (inert in bulk) becomes highly active for CO oxidation below 5 nm on TiO_2; Pt / Pd nanoparticles on carbon for fuel cells and automobile catalytic converters (reduces use of precious metal)
- Shape-selective catalysts (nano cubes, polyhedra expose particular facets), single atom catalysts, zeolites and mesoporous silica (MCM-41, SBA-15) with 2 – 50 nm pores for petroleum cracking
- Photocatalysts: TiO_2, ZnO, g-C_3N_4 produce electron-hole pairs → reactive radicals (·OH, O_2^-) for water splitting and pollutant degradation

4.3 Self-assembled monolayers (SAM) and surface engineering

- **SAMs**: thiols (R-SH) on Au, silanes (R-$SiCl_3$) on SiO_2. Molecules chemisorb and pack spontaneously in an ordered single layer (1 – 3 nm); end group controls wetting, adhesion, biocompatibility, corrosion resistance
- **Langmuir-Blodgett films** and layer-by-layer assembly build multilayers with nm control
- **Nano-patterning**: dip-pen nanolithography, micro-contact printing, nano-imprint for biochips, sensors, plasmonic substrates
- **Tribology**: nano-lubricants and DLC coatings reduce friction and wear (MEMS, hard disk, engines); **nano-indentation** and nano-scratching test surfaces
- **Sensors**: high surface area increases the adsorption signal (gas sensors from SnO_2, ZnO, WO_3 nanowires detect ppb levels); SERS substrates; QCM
- **Surface plasmon resonance (SPR)** biosensors measure binding of biomolecules by the refractive index change near a thin Au film

Surface effect	Nano-technique	Use

Water repellency	Nano roughness + low energy coating	Self-cleaning glass, fabrics
Adhesion control	Nano hairs, SAMs	Gecko tape, bio-adhesion
Reactivity	High surface area, nano facets	Catalysis, sensing, adsorption
Friction / wear	DLC, MoS_2 nanolayers	Engines, MEMS, hard disk

5. Applications in Energy

5.1 Solar energy

- **Third generation cells**: dye-sensitised solar cells (DSSC) use a mesoporous TiO_2 nano-layer (20 nm particles, 12 μm thick) with Ru dye; efficiency ~ 11 – 13 %
- **Quantum dot solar cells** (PbS, CdSe, $CuInS_2$): tunable band gap by size, **multiple exciton generation** (MEG: one high energy photon makes 2+ carriers) may exceed the Shockley-Queisser 33 % limit
- **Perovskite solar cells** ($CH_3NH_3PbI_3$ nano-crystalline films, > 26 % efficiency in labs); **thin-film CdTe, CIGS, a-Si**, organic (bulk-heterojunction) cells with nano-phase separated donor / acceptor
- **Light trapping**: Si nanowire / black silicon, plasmonic Au / Ag particles increase absorption in thin films; anti-reflection nano coatings

5.2 Energy storage

- **Li-ion batteries**: nano-$LiFePO_4$, Si nanowire / nanoparticle anodes (accommodate 300 % volume change, capacity 4200 mAh/g vs graphite 372), TiO_2 / $Li_4Ti_5O_{12}$ nano-anodes (fast charge), CNT / graphene conductive additives; shorter Li-ion diffusion path in nano-particles gives high power
- **Supercapacitors**: high surface area activated carbon, graphene, CNT, MnO_2 / RuO_2 nano-oxides (pseudocapacitance) deliver high power density and 10^5 cycles
- **Hydrogen storage**: nano-structured Mg / MgH_2, metal-organic

frameworks (MOFs), CNT, graphene: fast adsorption / desorption kinetics

- **Fuel cells**: Pt nano-catalysts on carbon, nano-structured electrolytes (YSZ thin films) in SOFC, nafion nano-composite membrane

5.3 Energy conversion and saving

- **Thermoelectrics**: nano-structured Bi_2Te_3, PbTe, SiGe, Si nanowires – low lattice thermal conductivity, ZT > 2 for waste heat recovery
- **Piezoelectric nano-generators** (ZnO nanowires) harvest mechanical energy; triboelectric nano-generators for wearables
- **Photocatalytic water splitting** for H_2 (TiO_2, $BiVO_4$, Fe_2O_3); nano-catalysts for CO_2 reduction
- **LED lighting** (GaN nano-structures, QD-LED, QLED TVs) saves energy; **smart windows**, **aerogel** thermal insulation for buildings; lightweight nano-composites for vehicles reduce fuel consumption; efficient nano-catalysts in refineries; high strength wind turbine blades (CNT-epoxy)

6. Applications in Environment

Area	Nano solution
Water purification	Nano-filtration / reverse osmosis membranes, CNT and graphene oxide membranes (desalination), nano-Ag filters for bacteria, nano-zero valent iron (nZVI) for removing Cr^{6+}, As, Pb and chlorinated solvents, magnetic nano-adsorbent (Fe_3O_4) that can be recovered, nano-clays, TiO_2 photocatalysis of dyes and pathogens
Air purification	Nano-TiO_2 coated filters, nano-catalysts in automobile converters (Pt-Pd-Rh) and diesel soot oxidation, nano-fibre HEPA masks, electrostatic filtration, VOC and NO_x removal by photocatalytic paints
Sensing / monitoring	Nano gas sensors for CO, NO_2, H_2S, VOCs; heavy metal sensors with Au nanoparticles (colorimetric); biosensors for pathogens in water; SERS pesticide detection

Remediation of soil	nZVI injection in groundwater plumes; nano-bioremediation; nano-fertiliser and nano-pesticide for precision agriculture (reduce chemical use)
Green manufacturing / waste	Nano-catalysts lower energy and waste; recyclable nano-adsorbents; oil spill clean-up using magnetic / hydrophobic sponges (graphene sponge, CNT aerogel)
Climate	CO_2 capture by MOFs / zeolites / nano-amines; photocatalytic CO_2 conversion; lightweight materials to cut emissions

Caution:
Environmental risk: engineered nanoparticles may accumulate and be toxic to organisms; life cycle assessment and safe-by-design are required.

7. Applications of Nanostructured Thin Films

Nanostructured thin films have thickness or grain size / porosity in the nanometre range (single layer, multilayer, superlattice, nano-composite, nano-porous), prepared by PVD / CVD / ALD / sol-gel / spin-coating, etc.

Application	Film / structure	Function
Optical coatings	Multilayer SiO_2 / TiO_2 (quarter-wave stacks), nano-porous SiO_2	Anti-reflection, dielectric mirrors, filters, beam splitters, low-E and heat-reflecting glass, Bragg reflectors in VCSEL
Solar cells	a-Si:H, CdTe, CIGS, perovskite, TCO (ITO, AZO, FTO)	Absorber and transparent electrodes, thin and flexible
Microelectronics	High-K HfO_2 (ALD), Si_3N_4, Cu barrier TaN, ferroelectric PZT films (FeRAM)	Gate dielectric, interconnects, memory, capacitors
Magnetic storage	Co/Pt, Fe/Cr multilayers, spin valves, MTJ (MgO), CoCrPt-SiO_2 granular films	GMR / TMR read heads, hard disks (> 1 Tb/in²), MRAM

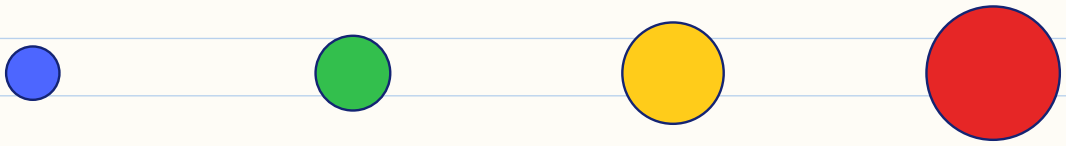
Sensors	SnO_2, ZnO, WO_3 nano-porous films; SAW devices (AlN, ZnO piezo films)	Gas, humidity, bio-sensing, MEMS
Hard / protective	TiN, TiAlN, CrN, DLC, nano-multilayers (TiN/AlN)	Cutting tools, wear and corrosion protection, decorative
Display and lighting	OLED organic multilayers, ITO, QLED, thin-film transistors (a-Si, IGZO)	Flat panel displays, flexible screens, solid state lighting
Energy storage	Thin-film Li batteries (LiPON electrolyte), electrochromic WO_3, supercapacitor films	Micro-batteries for IoT, smart windows
Superconducting	YBCO, Nb films, Josephson junctions	SQUID, qubits, RF filters
Thermoelectric	Bi_2Te_3 / Sb_2Te_3 superlattices	Micro-coolers, energy harvesting
Biomedical	Hydroxyapatite on implants, antibacterial Ag-TiO_2, DLC on stents	Biocompatibility, bone integration

- **Quantum wells and superlattices** (GaAs/AlGaAs) are the active layers in laser diodes, HEMT transistors, quantum cascade lasers, infrared photo-detectors
- **Multilayer films** utilise interference (optical), spin-dependent scattering (GMR) or phonon blocking (thermoelectric) effects only possible when layer thickness is in nm
- Dependence: properties depend on thickness, strain, interface, substrate and grain size; thin-film deposition control (ALD: 0.1 nm per cycle) is the Key

8. Applications of Quantum Dots

A quantum dot (QD) is a semiconductor nanocrystal (2 - 10 nm) in which carriers are confined in all three dimensions, so it has discrete atom-like energy levels and size-tunable band gap.

smaller dot → larger band gap → blue emission



CdSe 2 nm blue ~ 480 nm | CdSe 3 nm green ~ 530 | CdSe 4.5 nm yellow ~ 580 | CdSe 6 nm red ~ 630 nm

Fig: Size dependent colour of CdSe quantum dots (emission under UV)

Properties that make QDs useful

- Size-tunable absorption / emission, narrow emission (FWHM 20 - 30 nm), broad absorption, high quantum yield (> 90 %), photostability (100 - 1000 times more than organic dyes), long lifetime (10 - 30 ns), solution processable and low-cost
- Core-shell structure (CdSe/ZnS) reduces surface traps; non-toxic QDs: InP, $CuInS_2$, ZnSe, Si, graphene / carbon dots, perovskite ($CsPbBr_3$)

Field	Application of quantum dots
Displays	QLED / quantum-dot enhanced TVs (QD film on blue LED gives pure R, G, B, wider colour gamut > 100 % NTSC), electroluminescent QD-LED, micro-LED colour conversion
Lighting	White LEDs with QD phosphor: warm light, high CRI, lower energy; horticulture lighting
Solar cells	QD sensitised and QD solar cells (PbS, PbSe), luminescent solar concentrators (QD absorbs and re-emits to edge cells), MEG, tandem cells, intermediate band cells
Lasers and photonics	QD laser diodes (InAs/GaAs): low threshold current, temperature insensitive, used in optical communication; single photon source for quantum cryptography; optical amplifier
Photodetectors / imaging	PbS QD infrared detectors and cameras (SWIR), CMOS image sensors, night vision, thermal imaging
Bio-medical imaging	Fluorescent labels, in-vivo tumour imaging, FRET sensors, flow cytometry, multiplexed immunoassay, photodynamic therapy, drug tracking; image-guided surgery

Sensors	Fluorescence quenching sensors for pH, heavy metal, toxins, explosives; QD gas sensors
Electronics and computing	Single electron transistor (SET), quantum dot memory, QD cellular automata, spin / charge qubits in quantum computers (Loss-DiVincenzo)
Security / anti-counterfeiting	Invisible QD inks and tags (currency, documents), barcoding
Catalysis / environment	QD photocatalytic H_2 evolution, CO_2 reduction, dye degradation

- **Limitations**: toxicity of Cd / Pb based dots, blinking (intermittent fluorescence), aggregation and stability in water, cost of large-scale monodispersed production, regulation (RoHS)
- 2023 Nobel Prize in Chemistry (Bawendi, Brus, Ekimov) honours discovery and synthesis of quantum dots

Quick Revision - Important Questions

1. Discuss the applications of nanotechnology in material science (nanocomposites, coatings).
2. Explain nanomedicine: drug delivery, diagnostics and therapy with examples.
3. Explain the applications of nano-technology in surface science (lotus effect, catalysis, SAM).
4. How is nanotechnology useful in energy generation and storage? Discuss solar cells, batteries, supercapacitors.
5. Describe applications of nanomaterials in environmental protection (water and air purification).
6. Write about the applications of nanostructured thin films.
7. What are quantum dots? Write their properties and applications.